

F(AI)²R: Companion Technical Report

Appendices, Ladder Specification, Pipeline Prompts,
Provenance Analysis, Implementation Notes

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Abstract

This is the companion technical report to the F(AI)²R position paper . The position paper argues that FAIR is insufficient for LLM-assisted scholarly writing and proposes a multiplicative extension — F(AI)²R — enforced by eight integrated practices and a ten-stage agent pipeline; this companion carries the artefacts the argument leans on. Seven sections cover, in order, the verification-ladder specification used by the Source Analyzer and Aligner stages, the prompt taxonomy under agents/, the F(AI)²R cell-mapping crosswalk (FAIR sub-principles × AI passes), the per-step process-evolution chronology, the four diagnostic SPARQL queries on the live doc/provenance.ttl graph with worked numbers, the repository implementation (layout, CI, replay), and two forward-looking pattern extensions (domain ontologies and graph interconnectivity). The seven sections are referenced from the main paper via \companionref; their fair2r:Claim IRIs continue to resolve in the same shared doc/provenance.ttl graph, which spans both documents.

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1 Verification Ladder Specification

The ladder is the finite-state machine of the *verification ladder section*¹. This appendix supplies the per-rung specification used by the Source Analyzer (*pipeline section*²) and the Aligner.

Rung	Required artefact	Promotion condition	PRISMA / GRADE floor
UNVERIFIED-EXTERNAL	—	Claim recognised as needing a source.	—
NEEDS-RESEARCH	Search query in research-protocol has been identified.	A candidate URL or DOI has been identified.	—
LIT-RETRIEVED	URL/DOI that resolves	A Source Analyzer reads the source and confirms claim support, with quoted snippet.	PRISMA: Identification
AI-CONFIRMED	Quoted snippet in <code>sources.md</code> plus <code>fair2r:Source entity</code>	A human reads the source and confirms the inference from source to claim.	PRISMA: Screening; GRADE floor: LOW
LIT-READ	Vendored source in <code>doc/sources/</code> (when licensable) or human signature in logbook	—	PRISMA: Included; GRADE floor: MODERATE

Table 1. Verification ladder, with promotion condition and crosswalk floor. The crosswalk is the floor each rung permits, not a guarantee of the GRADE ceiling.

Retraction. A claim is retracted by adding a `prov:Invalidation` activity that names the original `fair2r:Claim` as `prov:invalidated`; the original triple is not deleted. A retracted claim cannot be invoked by the manuscript and is rendered visibly retracted on the public site.

Per-rung budget. At submission time, the manuscript reports the fraction of claims at each rung. A typical first submission carries a single-digit percentage at LIT-RETRIEVED, a moderate fraction at AI-CONFIRMED, and the load-bearing remainder at LIT-READ. The ratio is itself a disclosure under the sixth practice (*eight integrated practices*³, item 6).

2 Prompt Taxonomy

Every prompt under `agents/` follows the same five-section structure so that a different harness can ingest it with minimal adaptation.

Role. One paragraph: what the agent is, in plain language, with a one-sentence non-goal. The non-goal exists to refuse scope creep at read time.

Primary-artifact consistency (binding). The shared invariant of the *eight integrated practices*⁴, item 4, restated with a tailored clause for the agent’s role: which of (manuscript, graph,

¹See the section *verification ladder section* in the main F(AI)²R position paper [1].

²See the section *pipeline section* in the main F(AI)²R position paper [1].

³See the section *eight integrated practices* in the main F(AI)²R position paper [1].

⁴See the section *eight integrated practices* in the main F(AI)²R position paper [1].

logbook) the agent updates directly, which it submits for update by another agent, and what it must not finish without.

You do. The actions the agent is permitted to perform, in bullet form. Each bullet is small enough to be checked at review time.

You do not. The actions explicitly out of scope, in bullet form. The list is short and load-bearing: an agent that needs to be told “do not invent citations” is one prompt change away from doing so.

Output. The artefact the agent produces and where it lands. Outputs are typed: a draft is a diff, an audit is a defect registry, a graph update is a triple bundle, a section file is `paper/sections/<slug>.tex`.

Granularity rules.

- One agent per role; one role per file.
- One section per `.tex` file under `paper/sections/`; one `.tex` per section. Soft target: 100–400 lines per file. The `scientific-writer` prompt enforces this.
- One activity per stage in `doc/provenance.ttl`; an activity that runs more than once gets repeated entries with distinct `prov:startedAtTime`.

Refusal conditions. Each prompt names at least one refusal condition. Common ones: *do not invent citations*; *do not write to the graph if a submission lacks a corresponding manuscript change*; *do not deploy the public site without recorded publication consent*. Refusal is a feature, not a defect.

3 F(AI)²R Cell Mapping

The original FAIR Guiding Principles enumerate fifteen sub-principles. F(AI)²R adds, for each, an authoring-pass cell and an audit-pass cell that name the artefact involved. The full list is in `doc/fair.md`; Table 2 is a condensed view.

The table is a contract. Adopting F(AI)²R amounts to producing an audit pass artefact for every cell in the right column at submission time, and to declaring the residual gaps explicitly when an artefact does not yet exist.

4 Process evolution: the chronology

This appendix expands the digest table in the *process evolution section of the main paper*⁵ into a per-step chronology. The body of the paper carries the table because the table is what a reader needs at submission time; the chronology below is the longer record, kept here so the table is auditable against narrative without expanding the body.

Bootstrap. Researcher request: initialise a paper-plus-pipeline repository with prompts under `agents/`, a logbook under `doc/`, and a single PROV-O graph that follows the creation of the paper, distilled from a prior repository (`noheton/Obcurity-Is-Dead`) with all domain content abstracted away. The first AI draft read F(AI)²R as “AI participates twice in the loop”. Reading the prior repository carefully revealed the researcher’s original framing: (AI) as a factor multiplied through every FAIR axis, squared because of the two passes per axis. The paper draft adopts

⁵See the section *process evolution section of the main paper* in the main F(AI)²R position paper [1].

Sub-principle	Authoring pass (AI ₁)	Audit pass (AI ₂)
F1 (identifier)	Mint Zenodo DOI; assign fair2r:Claim IRIs.	Verify identifier resolves; verify IRIs are unique.
F2 (rich metadata)	Populate CFF / CodeMeta / Zenodo metadata.	Verify metadata surfaces agree.
F3 (metadata link)	Reference the data identifier from metadata.	Verify the link is bidirectional.
F4 (registered)	Deposit at Zenodo; minimum DataCite metadata.	Verify deposit visible from the registry.
A1 (retrievable)	Public Git repo; static site.	Verify clone-and-build succeeds on a clean machine.
A1.1 (open protocol)	HTTPS / Git.	—
A1.2 (auth where needed)	Document redaction policy and consent gate.	Verify redacted markers cover the documented taxonomy.
A2 (metadata persists)	Metadata in Git history; immutable Zenodo deposit.	Verify post-retraction metadata still reachable.
I1 (formal language)	Turtle for provenance; $\LaTeX$ for prose; CFF / JSON for metadata.	Validate Turtle; lint CFF.
I2 (FAIR vocab)	PROV-O + DCTERMS + FOAF + fair2r:.	Verify namespaces stable.
I3 (qualified refs)	prov:wasDerivedFrom, fair2r:contradicts, etc.	Verify graph has no dangling refs.
R1 (multiple attrs)	Record fair2r:VerificationState, prov:wasAttributedTo, etc.	Verify rung distribution at submission time.
R1.1 (clear licence)	MIT for code/RDF; CC-BY-4.0 for prose.	Verify SPDX expressions.
R1.2 (provenance)	The provenance graph itself.	Replay the graph; report drift.
R1.3 (community std)	Follow FAIR4RS / FAIR4ML where applicable.	Cross-check against the relevant standard at audit time.

Table 2. F(AI)²R cell mapping. Authoritative source: `doc/fair.md` in the public repository.

the latter. Lesson: when the AI's first-pass interpretation differs from the researcher's prior writing, the prior writing wins.

Primary-artefact consistency invariant. After the bootstrap, the researcher asserted that the manuscript, the PROV-O graph, and the logbook are primary artefacts and must remain consistent and up to date at all times. The AI added a binding “Primary-artefact consistency” block to every prompt; named the Orchestrator the ultimate custodian and the Provenance Curator the keystone; specified that the FAIR Aligner surfaces a desync as an audit *fail*. The rule was recorded as a fair2r:Claim, attributed to the researcher, and as a prov:Activity revising every affected prompt entity.

Static site, Pages workflow, paper-build CI. Researcher request: a public site in the spirit of the prior repository, plus continuous integration. The AI added a static-site generator (parsing `doc/provenance.ttl` via `rdflib`), the corresponding GitHub Pages workflow with a Turtle-validation gate, and a paper-build workflow.

Chapter granularity for the writer. Researcher request: *each .tex file should represent one chapter*. The Scientific Writer prompt gained a binding “File granularity” block: one `\section{}` per file, soft target 100–400 lines, filename slug matches the `\label{sec:...}`. `paper/main.tex` became a thin assembler.

Recursive attestation, by request. Researcher request: *add a section that this paper is generated using exactly this process*. A dedicated *self-reference* section (`paper/sections/self-reference.tex`) was added at the time, then retired in a later trim pass after the same attestation had been absorbed into the *Author's Note in the main paper*⁶ and into practice 5 of the *eight integrated practices*⁷; `prov:Invalidation` on the section is recorded in `doc/provenance.ttl`.

This section, by request. Researcher request: *document the evolution of the process during cooperative writing*. This section. The `fair2r:Claim` entries for it sit at AI-CONFIRMED until reviewed.

Short-form pivot. Researcher request: write only the short form (10 pages of body plus references and appendices) rather than maintaining a long/condensed split. The `main-condensed.tex` target was retired; the Condenser agent was repurposed as a *page-budget enforcer* that fires when the compiled PDF exceeds the configured budget; the Scientific Writer prompt gained a page-budget rule.

Institutional design system. Researcher request: integrate the DLR design system (Frutiger / Arial, blue/green/yellow chapter accents, square corners, hairline borders, photography-led, no marketing register). The site CSS was rewritten on top of the DLR `colors_and_type.css` tokens; a small $\text{\LaTeX}$ style file (`paper/style/fair2r.sty`) carries the equivalent typography and rule colours into the PDF; the imprint was updated with the author's institutional identity (DLR ZLP Augsburg, Helmholtz, NFDI4Ing, HMC).

Patterns observed

Three patterns recur and are worth naming.

Researcher-as-corrector. The researcher's interventions are short, specific, and rule-shaped. None is a rewrite of AI-produced prose. The cooperation lives in the gap between a model that produces a plausible default and a researcher who can recognise when the default is wrong.

Pipeline as target, not yet as practice. The ten-stage pipeline is the architecture the prompts encode. The present draft was produced by a single AI agent in conversation with the researcher, with role separation documented in the prompts but not yet actively switched per turn. We name this gap rather than paper over it.

Logbook as conscience. Every event above corresponds to an entry in `doc/logbook.md`. The append-only discipline made the reconstruction cheap. *The logbook is the conscience of the cooperation; the provenance graph is its record.*

Field notes from the second cooperative writing project

The methodological ancestor [2] was the author's first cooperative-writing project. The present paper is the second. Three observations from the second-time experience are worth recording; the full set lives in `doc/user-observations-log.md`.

The researcher's role is shifting toward experimentation. Cooperative writing with an LLM, as practised here, lifts the prose-production load off the researcher and leaves more cycles for abstraction, decision-making, and design of the next experiment. For empirical disciplines

⁶See the section *Author's Note in the main paper* in the main F(AI)²R position paper [1].

⁷See the section *eight integrated practices* in the main F(AI)²R position paper [1].

this is a net good: writing-as-administrative- overhead shrinks; experimentation-as-the-actual-contribution grows. For paper-driven disciplines — philosophy, parts of the humanities — the picture is more uncomfortable: the prose *is* the experiment, and removing the slow, deliberate craft of writing risks removing the thinking that the writing was doing [3]

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. We name the asymmetry rather than pretend F(AI)²R is discipline-neutral.

Prompt-craft is closer to programming than to drafting. A second-time observation: writing a prompt feels easier than writing the corresponding paragraph oneself. The short conversational cycle maps, for a researcher trained in computer science, to the familiar loop of writing a program and watching its output — closer to experimentation than to formal academic writing [4, 5]

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. This is a feature (it lowers the activation energy of getting started, which is itself a contribution to throughput) and a risk (prompt-craft can become a comfortable substitute for the harder parts of writing). The risk is managed, in F(AI)²R, by keeping the human in the deciding seat (see the *Author's Note*⁸ subsection *What surprised me about the cooperation* in the main paper) and by the per-claim provenance map that makes drift detectable.

Responsibility includes the rooms after publication. The model-side contribution ends at the commit. The human-side contribution continues into the rooms where the work has to be defended in person — conferences, panels, discussions, reviewer correspondence. AI cannot share that load: the model cannot consent, cannot carry a reputation, and cannot be cross-examined at the next RDA plenary or HMC symposium. The seventh integrated practice (*eight integrated practices*⁹, item 7) names the legal version of this point; the field-notes version is more ordinary. *Whoever puts their name on the paper signs up for every conversation the paper produces afterwards*, and the AI's contribution is, by construction, absent from those conversations. The interpersonal-sharing leg of authorship is the leg the methodology section cannot eliminate even in principle.

5 Provenance analysis: queries and worked numbers

The *provenance-analysis section of the main paper*¹⁰ summarises the audit; this appendix walks the four diagnostic queries on the live graph. The auto-generated tables below are produced by `scripts/provenance_analysis.py`, which the build pipeline re-runs on every commit. They reflect the state of `doc/provenance.ttl` at the time this PDF was compiled and are embedded into the prose by the build, not transcribed by hand.

Verification rung distribution. Every claim in the graph carries a `fair2r:verificationState`; we report the fraction at each rung at submission time per the sixth integrated practice (*eight integrated practices*¹¹, item 6). The query is one SPARQL pattern:

```
SELECT ?state (COUNT(?c) AS ?n)
WHERE { ?c a fair2r:Claim ; fair2r:verificationState ?state }
GROUP BY ?state
```

The result for the present manuscript:

⁸See the section *Author's Note* in the main F(AI)²R position paper [1].

⁹See the section *eight integrated practices* in the main F(AI)²R position paper [1].

¹⁰See the section *provenance-analysis section of the main paper* in the main F(AI)²R position paper [1].

¹¹See the section *eight integrated practices* in the main F(AI)²R position paper [1].

Rung	Claims	Share
needs-research	2	4.7 %
ai-confirmed	7	16.3 %
human-confirmed	24	55.8 %
source-vendored	9	20.9 %
human-read	1	2.3 %
Total	43	

Section-to-claim coverage. Every section in `paper/main.tex` should appear in the graph as a `fair2r:Section` entity, with at least one `prov:Activity` that produced it and (almost always) at least one `fair2r:Claim` attributed to that activity. A section that exists in prose but not in the graph is a defect surfaced by the FAIR Aligner.

Section heading	Label	In graph?	Claims
Verification Ladder Specification	<code>sec:appendix-ladder</code>	no	0
Prompt Taxonomy	<code>sec:appendix-prompts</code>	no	0
F(AI) ²	<code>sec:appendix-cell-mapping</code>	no	0
Process evolution: the chronology	<code>sec:appendix-evolution</code>	no	0
Provenance analysis: queries and worked numbers	<code>sec:appendix-provenance-analysis</code>	yes	0
Implementation: layout, CI, replay	<code>sec:appendix-implementation</code>	yes	0
Pattern extensions: domain ontologies and graph-as-research-infrastructure	<code>sec:appendix-pattern-extensions</code>	yes	0
Author's Note — Advice for reading	<code>sec:authors-note</code>	yes	6
Conclusion	<code>sec:conclusion</code>	yes	11
Eight Integrated Practices	<code>sec:eight</code>	yes	10
Process Evolution During Cooperative Writing	<code>sec:evolution</code>	yes	10
Failure Modes Named and Addressed	<code>sec:failure-modes</code>	yes	10
Introduction	<code>sec:intro</code>	yes	3
Limits, dual-use, and standing objections	<code>sec:limits-and-objectives</code>	yes	0
The F(AI) ² Position	<code>sec:pattern</code>	yes	10
Provenance Analysis	<code>sec:position</code>	yes	6
Statement of authorship and AI use	<code>sec:provenance-analysis</code>	yes	2
	<code>sec:authorship-statement</code>	yes	6

A coverage row that reads *prose: yes / graph: no* is a hard fail; *graph: yes / claims: 0* is a soft warning meaning the section has been registered but the claim curation has not caught up.

Source invocation strength. Every `fair2r:Source` entity that the manuscript cites should appear as the object of at least one `prov:wasDerivedFrom` edge from a `fair2r:Claim`. A source cited but never linked from a claim is most often a **ghost citation** — a reference that lives in

the bibliography because the prose mentions it, but that no claim actually relies on. In a healthy F(AI)²R manuscript, the count is zero.

Sources cited: 56; invoked by at least one claim: 28; **ghost citations: 28.**

Source	Title	Invocations
aclrr_llm_policy	aclrr_llm_policy	1
alemohammad2023mad	alemohammad2023mad	0
anderson2024homogenization	anderson2024homogenization	0
ashburner2000go	ashburner2000go	2
bender2021stochastic	bender2021stochastic	0
benjelloun2024croissant	benjelloun2024croissant	1
birhane2022values	birhane2022values	0
chen2023drift	chen2023drift	0
chuehong2022fair4rs	chuehong2022fair4rs	0
clark1998extended	clark1998extended	0
clark2025extending	clark2025extending	0
clarke2009modelchecking	clarke2009modelchecking	1
conroy2023sleuths	conroy2023sleuths	1
curdt2025hmc	curdt2025hmc	1
eisen2018preprints	eisen2018preprints	1
else2023chatgpt	else2023chatgpt	1
gebru2021datasheets	gebru2021datasheets	0
guyatt2008grade	guyatt2008grade	0
hutchins1995cognition	hutchins1995cognition	0
iclr_llm_policy	iclr_llm_policy	1
icmje2023	icmje2023	1
ioannidis2005	ioannidis2005	1
janowicz2019sosa	janowicz2019sosa	1
klein2009sel4	klein2009sel4	1
kobak2024delving	kobak2024delving	1
krebs2026obscurity	krebs2026obscurity	10
kuteeva2024diversity	kuteeva2024diversity	0
li2023thirsty	li2023thirsty	0
liang2024mapping	liang2024mapping	1
liu2023prompt	liu2023prompt	0
liu2026ara	liu2026ara	1
luccioni2024power	luccioni2024power	0
magesh2024legal	magesh2024legal	2
mitchell2019modelcards	mitchell2019modelcards	0
neurips_llm_policy	neurips_llm_policy	1
page2021prisma	page2021prisma	0
patterson2021carbon	patterson2021carbon	0
pineau2021reproducibility	pineau2021reproducibility	1
qudt	qudt	1
ravi2024fair4ml	ravi2024fair4ml	0
reynolds2021prompt	reynolds2021prompt	0
rijgersberg2013om	rijgersberg2013om	1
sadasivan2023reliably	sadasivan2023reliably	0
schmitt2020nfdi4ing	schmitt2020nfdi4ing	0
shafer2014xennials	shafer2014xennials	1
shumailov2024collapse	shumailov2024collapse	0
strubell2019energy	strubell2019energy	0
tennant2016open	tennant2016open	1
thorp2023chatgpt	thorp2023chatgpt	0
urhg2	urhg2	0
usco2023ai	usco2023ai	0
vannoorden2023chatgpt	vannoorden2023chatgpt	1
vaswani2017attention	vaswani2017attention	1
w3c2013provo	w3c2013provo	0
walters2023fabrication	walters2023fabrication	2
williams2013of	williams2013of	1

Activity timeline. The `prov:Activity` nodes form a dated record of every substantive change. The most recent five activities, in ISO-8601 order, are:

Date	Activity	Label
2026-05-13	integrate-external-works	Integrate 5-13 three external works (Ara, Vaswani, Croissant) into the manuscript
2026-05-08	provenance-graph-hygiene	Provenance graph hygiene pass after audit
2026-05-08	fix-poster-density	Fix portrait poster vertical density; add density rule to layout-scrutinizer
2026-05-08	expand-layout-scrutinizer	Expand layout-scrutinizer scope to all five primary artefacts; flip poster from
2026-05-07	write-provenance-verification	Author the provenance-verification scoping note

The same data, with one row per session, is the human-readable mirror in `doc/logbook.md`; drift between the two is itself a defect (see the *handback-discipline subsection of the main paper*¹²).

Keep this appendix honest. This appendix is generated against the current graph. If a future revision adds a verification rung, a new section, or a new failure mode, the corresponding entry here updates automatically on the next build; the prose around the figures stays put. The intent is that a reader who clones the repository, recompiles the PDF, and re-reads this appendix, will see the analysis reflect the manuscript that was actually produced — not a snapshot frozen at draft time.

6 Implementation: layout, CI, replay

The pattern is materialised in the public repository that produced this paper.¹³ The intent is not to prescribe a stack: any version-control system with a build target, a structured-data format that supports PROV-O, and a way to version prompt files would do.

Layout. `paper/` carries the $\text{\LaTeX}$ sources, bibliography, Makefile, and figure sources, with one `\section{}` per file under `paper/sections/`. `doc/` carries the methodology, FAIR notes, human-AI collaboration policy, research protocol, submission plan, the append-only logbook, the PROV-O graph in Turtle, and a Mermaid mirror. `agents/` carries one markdown file per role (orchestrator, scientific writer, source analyzer, FAIR aligner, provenance curator, layout scrutinizer, readability reviewer, illustration, condenser, research protocol), each structured per the prompt taxonomy of Appendix 2. `scripts/` carries the static-site generator that parses `doc/provenance.ttl` via `rdflib` and renders it into a small HTML site.

Continuous integration. `build-paper.yml` compiles the manuscript on every push that touches `paper/` and uploads the PDF as an artefact, using `xu-cheng/latex-action@v3`. `pages.yml` validates the Turtle, builds the site, and (on `main`) deploys via the official GitHub Pages action chain. Working branches see the build output as a downloadable artefact, never as a public deploy.

Replay loop. A reader who clones the repository, installs a TeX Live distribution and the dependencies in `scripts/requirements.txt`, runs `make -C paper pdf` and `python scripts/build_provenance_si`

¹²See the section *handback-discipline subsection of the main paper* in the main F(AI)²R position paper [1].

¹³<https://github.com/noheton/f-ai-r>

has reproduced the paper, the provenance graph, and the public site. The audit pass for **Reusable** is, in our setup, that loop run on a clean machine, compared against the published artefact. Any difference is a defect.

7 Pattern extensions: domain ontologies and graph-as-research-infrastructure

This appendix expands two forward-looking subsections that the *pattern section of the main paper*¹⁴ points at: how the pattern composes with domain-specific ontologies, and how the bioinformatics precedent for semantic-tagging-as-norm reframes provenance graphs as research infrastructure under volume pressure. Neither is on the critical path of the eight integrated practices, but each is how F(AI)²R generalises beyond the present case.

7.1 Extensibility through domain ontologies

PROV-O is the spine; it does not have to be the only ontology in the graph. The schema we have introduced is compositional: any `fair2r:Claim` can carry additional triples that ground its *values* in a domain ontology, alongside the `prov:wasDerivedFrom` edge that grounds its *evidence*. The two axes are orthogonal — provenance asks *where did this claim come from?*, domain ontologies ask *what does the value of this claim mean?* — and a manuscript that answers both is more reproducible than one that answers only the first.

A neighbouring vocabulary worth naming explicitly is MLCommons Croissant 1.0 [6], the schema.org / JSON-LD vocabulary for ML datasets and their responsible-AI annotations. Croissant occupies the dataset-metadata slot (see the *limits and objections section of the main paper*¹⁵) and composes cleanly with the F(AI)²R schema: a `fair2r:Source` can reference a Croissant record so that the provenance graph for a manuscript and the dataset metadata for the data it sits on top of read as a single linked-data graph rather than two disconnected ones.

Two examples concrete enough to plug in today:

Sensor data: SOSA / SSN. The W3C Semantic Sensor Network ontology and its lightweight core SOSA [7]

verify

expose `sosa:Observation`, `sosa:Sensor`, `sosa:FeatureOfInterest`, and the relations between them. A `fair2r:Claim` that asserts a measurement can carry a `prov:wasDerivedFrom` pointing at the underlying `sosa:Observation`, and the audit pass gains the ability to ask *is this measurement well-formed?* as a SHACL query rather than a paragraph of prose.

Units of measure: OM-2 / QUDT. The OM-2 ontology of units of measure [8]

verify

and the QUDT vocabulary [9]

verify

expose `om:Quantity`, `om:Unit`, `qudt:hasUnit`, and the dimensional relations between units. A claim that asserts a numeric value can carry the unit as an explicit triple rather than as a typographic decoration. Unit consistency is then a graph query, not a manuscript-reading exercise.

¹⁴See the section *pattern section of the main paper* in the main F(AI)²R position paper [1].

¹⁵See the section *limits and objections section of the main paper* in the main F(AI)²R position paper [1].

The class of failure modes this addresses is the one prose alone cannot reach: a value that is internally coherent in its prose and its citation but *wrong in its units*, or *wrong in the dimensional sense it implies*. A reviewer catching such a defect today reads the figure caption and the methods section and triangulates; an audit pass with a domain ontology reads the graph and the SHACL shapes and reports.

The recommendation is open. F(AI)²R does not prescribe specific domain ontologies because the right choice is discipline-specific. It prescribes that, where a discipline already has a *konkrete standardisierte Methode* expressed as an ontology, the F(AI)²R repository should adopt it and link from `fair2r:Claim` entities to its terms. NFDI4Ing and the Helmholtz Metadata Collaboration are the nearest two communities to this paper that maintain such ontologies for engineering and metadata respectively, and the integration cost is low. The ceiling is the discipline's; the floor is what F(AI)²R asks for.

7.2 Looking forward: provenance graphs as research infrastructure

The argument of §7.1 generalises. The discipline that pioneered semantic-tagging-as-norm under data-volume pressure is bioinformatics: faced with what was, at the time, an obscene amount of sequence data, the field standardised identifier and annotation schemas (the Gene Ontology [10])

verify

being the most visible example) so that genes, proteins, and their relations could be referenced unambiguously across labs, papers, and databases. The ontology was not a research output; it was a precondition for research that scaled. The community's investment in semantic tagging was, in retrospect, what made the next decade of bioinformatics possible.

We claim a closer analogy than is sometimes recognised. LLM-assisted scholarly writing is producing artefacts at a volume that the unaided reviewer pipeline cannot absorb (see the *introduction of the main paper*¹⁶). The response will, on present trajectory, look more like the bioinformatics response than like the editorial response: not new detection tools but new shared schemas, ontologies, and graph-interconnected metadata that let claims, sources, and activities reference each other across institutional boundaries. **Future research success may come to depend less on the solitary craft of a single manuscript and more on three things the F(AI)²R schema is designed to support:** (i) provenance models that turn published claims into machine-readable hypotheses for the next study; (ii) graph interconnectivity across institutional knowledge graphs — the Helmholtz cross-centre graph [11] is the proximate example, and a fully-instrumented version would let a claim in one centre's manuscript be linked, queried, and challenged by claims in another's without leaving the linked-data layer; (iii) clever experimental validation that uses the graph to identify the next testable deviation from the published consensus.

That trajectory is a position the paper is willing to take and a direction the eight integrated practices serve. The verification ladder feeds (i); the per-claim provenance map and the domain-ontology side-arms feed (ii); the recursive case study and the audit pass feed (iii). None of this requires F(AI)²R to be the standard. It requires only that some standard with these properties be adopted, and the present paper is one offer among several the community can refine.

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¹⁶See the section *introduction of the main paper* in the main F(AI)²R position paper [1].

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